Hindi–English simultaneous translation: research, data design, entity checking and a wait-k study

[Your Name] · Inter IIT Tech Meet 15.0, Sarvam.ai mini PS, Parts B–E (ML track) · code + RESULTS.md in the repository

Status of evidence. Recordings made: **0/6**; transcripts verified against audio: **0/6**; timing source: **placeholder_estimate**. All second-based latency numbers below come from **placeholder timings estimated from text, not from audio**, and the alignment audit is pending. Word-based AL and evidence-safe rates depend only on transcripts and translations. Throughout, the wait-k experiment is a **timing simulation over a fixed reference translation**; it does not reproduce the errors of a real streaming model generating from a prefix.

Part B: Four research proposals and one original idea

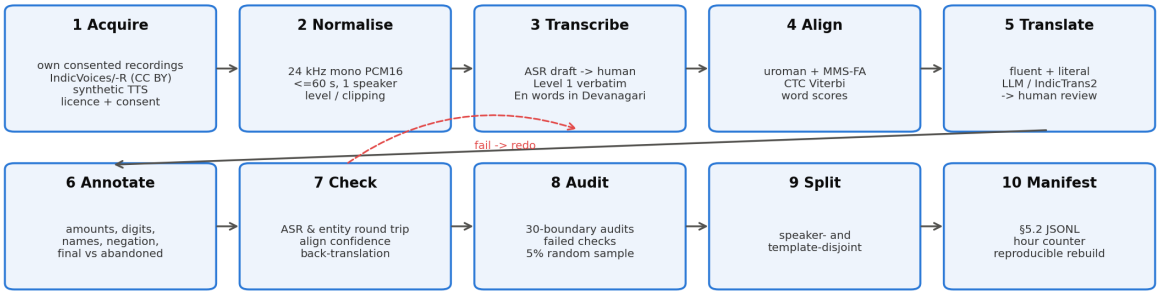
Question	Recommendation and evidence	Main risk / cheapest experiment	Unanswered question · claim checked
1. Timing the English (training targets, Q2+Q3)	Sentence-level fluent targets kept close to Hindi clause order. Per-word earliest-safe time = Hibiki contextual alignment (argmax jump in MT log-likelihood as the source prefix grows, Eq. 6 [4]) *and* an entity floor (never before the Hindi word that fixes the entity), plus a tuned safety lag. Hibiki Table 4: constant 2 s lag gives ASR-BLEU 10.0; sentence alignment 25.6 at LAAL 21.5 s; contextual alignment 26.6 at 5.0 s [4].	Wrong alignment teaches speaking before the deciding verb (band karva diya) and cannot be undone. Experiment: on our 6 clips, count entity words that the LM-jump alignment places before their determining Hindi word.	Safety lag when the deciding word is an English verb written in Devanagari? · *'Hibiki sets delays by perplexity'* (our draft notes): it uses the per-word *increase* in log-likelihood plus a fixed 2 s lag [4 §3.2].
2. Hard content (Q4)	Entity-rich subset from the IndicVoices Hindi prompt banks (digital transactions, customer care, named entities [5]) as minimal pairs differing in one fact; structured entity annotations; TTS verified by ASR round trip + deterministic Hindi number parsing on the entity span. Sarvam FAQ: 'speaking the abandoned value as fact is wrong' [1].	TTS says lakh where the script says crore; text-only checks pass. Experiment: 50 minimal pairs × 2 voices → ASR → parser mismatch rate.	How to score 'five lakh, sorry, five crore'? · *'Fluent = accurate'*: our clip02 fluent reference is only producible after waiting a full sentence; an equally fluent version with the abandoned digit is wrong.
3. Data sources, real vs synthetic (Q1)	Real-first: ~80 h consented code-mixed + ~40 h CC BY 4.0 IndicVoices/-R filtered for English words + ~60 h synthetic + ~20 h entity-rich tag (inside the others). Hibiki: ~40K h synthetic then ~900 h fine-tune, Fr–En only [4].	Model fits clean scripted speech, fails on real telephone code-mixing. Experiment: equal-hours real / synthetic / mixed, tested on speaker-disjoint human recordings.	Hindi hours with English words in IndicVoices? Not on any card [5,6]. · *'IndicVoices has 12,000 h'*: README 12,000/3,200 transcribed, arXiv 7,348/1,639, HF card 23.7K/11.2K [5]: quote the release.
4. Latency decoding choices (Q6)	Always report AL in source words (STACL Eq. 9 [2]), AL/LAAL in playback seconds (SimulEval [3], LAAL [7]) and End Offset [4]. Base on wait-k [2]; add entity-aware holding (below).	A measurement bug flatters a system (decision time instead of playback time). Experiment: this paper's simulator, same text, vary only the policy.	Seconds of lag worth one avoided entity error under relative scoring? · *'The SimulEval paper defines AL'* (mini PS): AL is defined in STACL §4.2 Eqs. 8–9 [2]; SimulEval implements it on time delays [3].

Original idea: entity-aware adaptive waiting (EAW), a hypothesis. Most English words (greetings, 'my account number is') are safe early; only commitments that cannot be retracted (amounts, digits, names, negation, the value after a correction) need evidence. **Rule (causal, implemented):** on top of wait-k, before committing a *risky English word* (number, scale, negation cue, non-initial capitalised word), keep reading while any of the last 2 heard Hindi words is risky (number/scale word, nahin/mat, correction marker such as sorry/matlab/are); variant *EAW-pause* also waits for a ≥ 0.3 s pause; cap 6 extra words. **Why here:** Hindi puts the scale after the number, corrections after a hesitation, negation at the clause end. **Ablation:** lexical vs pause trigger; window 1–3. **Falsified if** no EAW point lies above the wait-k curve at equal playback-time AL on real timings. **Fails** where the English must commit polarity before the Hindi negation is heard (Part E).

Part C: Dataset design for ≤ 200 h (a proposal, nothing collected)

Source	Licence / access (checked 15 Oct 2026)	Hours (as published)	Code-mixed?	Role
IndicVoices	CC BY 4.0, HF-gated	23.7K total / 11.2K transcribed (card); Hindi not stated	Not documented: measure	40 h pool after English-word filter
IndicVoices-R	CC BY 4.0, gated	1,704 h, 9–175 h/lang, 48 kHz	Same unknown	Clean speech; Indic TTS training
Kathbath	Card inconsistent: cc-by-4.0 vs CC0 text	1,684 h / 12 langs	Not verified	Low priority until licence resolved
Own consented recordings	Our consent form (covers sharing with Sarvam)	Budget: 80 h	Yes, by design	Natural code-mixing + entities
Synthetic (Bulbul hi / open TTS)	Sarvam outputs: competition use only, no redistribution [1 §10]	Each file counts (60 h)	Text-controlled	Rare entities, corrections
Aligners	MMS-FA HF port: CC-BY-NC-4.0 [8]; WhisperX Hindi default: card WER 72.6%, no licence [9]; torchaudio tutorial: English-only model [10]			We use MMS-FA + own CTC Viterbi

Accounting. Real audio counts as the union of source intervals; every synthetic file counts, including the same text in another voice; augmentation (noise, 300–3400 Hz telephone band, low-bitrate codec, speed) is applied at training time and adds 0 h [1 §3]. The entity-rich 20 h is a tag on records inside the real and synthetic pools, never a second copy. **Transcripts:** ASR draft, then a human edits it to IndicVoices Level 1 ('transcribed exactly as spoken' [5 §6]: fillers, repeats, English in Devanagari, numbers as words). **Timing:** uroman romanisation + MMS-FA; the Devanagari is preserved and the romanisation stored separately (schwa retention, apna→'apanaa', makes word *ends* less reliable than onsets).



Check (thresholds are initial engineering choices, to calibrate)	Threshold
Format 24 kHz/mono/PCM16, ≤ 60 s, Level-1 text (no Latin, no digits, sentence-final punctuation)	hard
Level: RMS ≥ –40 dBFS, clipping ≤ 0.1%, ≤ 50% near-silent frames	reject
ASR round-trip CER (synthetic / real)	≤ 15% / ≤ 30%
Entity round trip: numbers and digits parsed from the ASR of the entity span = annotation	exact, else regenerate

Check (thresholds are initial engineering choices, to calibrate)	Threshold
Alignment: mean word score; share of words with score < 0.3; 30-boundary onset audit per batch	≥ 0.5; ≤ 10%; median ≤ 50 ms
Back-translation chrF and entity checker on the English	≥ 40 and all entities pass
Speaker and template overlap across splits	0

Failure modes, including ones that pass the audit: every correction follows one template ('nahin nahin, X'), so the model learns the marker and not the meaning; fluent references put entities ahead of their Hindi evidence, which teaches the model to anticipate amounts; TTS prosody dominates the hours; speaker or template leakage inflates test scores; romanisation shifts timing targets. **Synthetic–real gap** = metric difference between matched synthetic and human-recorded held-out sets, tracked as synthetic hours grow. **Three questions to test first:** (1) share of IndicVoices Hindi utterances with English words and how they are scripted; (2) does synthetic entity-rich speech raise entity accuracy on *real* held-out clips; (3) is MMS-FA onset error on code-mixed Devanagari ≤ 50 ms median.

Part D: Six clips, transcripts, alignment and the entity checker

Clip	Content	Entities	Audio	Transcript
clip01	business loan enquiry; lakh -> crore self-correction	name×3, correction×1, negation×1	pending_recording	scripted_unverified
clip02	phone number dictation with a last-digit correction	correction×1, digits×1, name×1, negation×1	pending_recording	scripted_unverified
clip03	disputed card payment; several negations and a prohibition	negation×5, amount×1	pending_recording	scripted_unverified
clip04	meeting with named people; time self-correction; crore budget	name×5, correction×1, amount×1, negation×1	pending_recording	scripted_unverified
clip05	flat purchase; fractional lakh/crore amounts	name×1, amount×4, negation×2	pending_recording	scripted_unverified
clip06	bank transfer limit; account number; thousand -> lakh self-correction	name×3, digits×1, correction×1, amount×1, negation×1	pending_recording	scripted_unverified

Recording and transcription. Prompted, unscripted phone-style speech covering lakh/crore (incl. dedh, dhai, saadhe), a phone number with a last-digit correction, an account number, 13 names/organisations/places, 10 negations (incl. mat) + 1 affirmative control, and 4 self-corrections (amount ×2, digit, time). Transcripts follow Sarvam/IndicVoices Level 1 and are validated automatically (no Latin, no digits, final punctuation; Hindi amounts must parse to the annotated values). The audio is the source of truth: a transcript is marked *verbatim_verified* only after listening. **Aligner:** uroman (lcode=hin) → MMS-300m emissions (20 ms frames) → our numpy CTC Viterbi over the blank-interleaved label sequence (the tutorial's trellis, with proper blank states); on CPU it takes about 1.6 s for 6 s of audio. The English-only tutorial model cannot read Hindi. **Alignment audit: PENDING** (needs recordings): 30 word onsets (5/clip, including the determining words of entities), marked blind in Audacity before viewing automatic labels; we report mean, median and max |error| and counts over 50/100 ms.

Entity checker. Plain Python: Hindi/English number grammars (0–99 Hindi words, sau/hazaar/lakh/crore, fractions; English words, Indian and Western digit grouping, 'and a half', k/cr), digit-run extraction (double/triple, sentence boundaries), time parsing, a clause-scoped negation test (cue within 4 words before the predicate, no clause break, 'not only' excluded), and correction logic (final value present; abandoned value absent unless explicitly rejected, 'not X'). Unparseable values fail; nothing unknown passes. **Evaluation:** 66 hand-labelled variants of the fluent references (paraphrases, unit and digit forms, corruptions; labels fixed before running) = 415 decisions. Accuracy **98%**; error recall 100% (26/26); error precision 74% (9 false alarms). On non-'hard' variants 100%; on the 'hard' paraphrases 86%.

Category	Decisions	Accuracy	False alarms	Missed errors
amount	79	100%	0	0
correction	44	91%	4	0
digits	20	100%	0	0
name	151	100%	0	0
negation	121	96%	5	0

Where it fails (all 9 errors are false alarms on paraphrases): negation not expressed as a cue before the verb ('made *no such* payment', '*unless* it comes back', '*unable* to come', '*unlikely* to come', '*rather than* mine'); a retracted abandoned value ('five lakh, *sorry*, five crore'), which our labels accept but the checker rejects; the time '4.30' read as the amount 4.3. On literal glosses 7/37 checks fail because Hindi order puts 'not' after the verb. Development note: the variant set exposed one real bug (digit runs merging across a sentence end) which we fixed, so the set is not fully held out. The checker measures references and simulated outputs; it says nothing about a translation model, and its gold labels are the author's own.

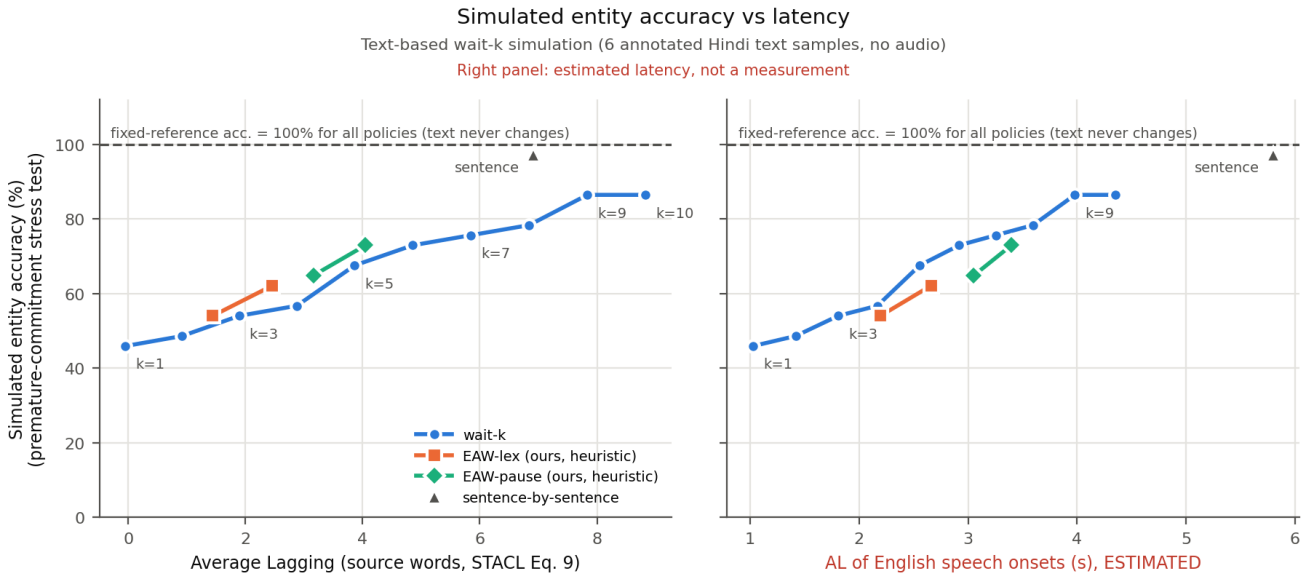
Part E (ML): wait-k on the clip timelines; where should a streaming model wait?

Setup. Source word j is 'heard' at its aligned end time. g(t) = Hindi words read when English word t is committed; wait-k: g(t) = min(k+t–1, |x|) [2] (= the mini PS 'k + i' with 0-based i); the last word waits for the end of input. Decision time D_t = end of word g(t) (+0.3 s if a pause was needed); each English word is spoken for a fixed 0.40 s (≈150 wpm, an estimate) and starts at max(D_t, end of the previous word), Sarvam's playback rule [1 §5.1]. We report AL in source words (STACL Eq. 9, τ = first t with g(t)=|x|, r=|y|/|x|), SimulEval AL on decision times and on speech onsets (seconds; = LAAL here because output = reference [3,7]), first-output time and End Offset. **Entity scoring.** The fixed reference passes 100% for every policy, which only says the reference is right. We therefore add a transparent stress test: each of the 37 entities has a hand-annotated determining Hindi word and a hand-written premature guess (e.g. 'five lakh'); if the policy commits the English entity before that word is heard, the guess replaces it and the checker scores the result. Unit tests verify the hand-computed Part A example (wait-3 on 'maine kal apna purana bank khata band karva diya': g = 3,4,5,6,7,8,9,9; τ = 7; AL = 2.625 words).

Results (6 clips, means; entities pooled) (placeholder timings: seconds columns are NOT measurements)

Policy	AL words	AL decision s	AL speech s	First out s	End offset s	Entities safe = simulated acc.
wait-1	-0.05	0.26	1.03	0.59	2.46	46%
wait-2	0.92	0.63	1.42	1.03	2.72	49%
wait-3	1.89	1.00	1.81	1.42	3.02	54%
wait-4	2.88	1.38	2.17	1.78	3.33	57%
wait-5	3.86	1.76	2.56	2.21	3.65	68%
wait-6	4.86	2.15	2.92	2.55	3.95	73%
wait-7	5.85	2.53	3.26	2.85	4.25	76%
wait-8	6.84	2.92	3.60	3.22	4.56	78%
wait-9	7.84	3.30	3.98	3.63	4.89	86%
wait-10	8.83	3.69	4.35	4.01	5.21	86%
EAW-lex k=2	1.44	0.85	2.20	1.03	3.57	54%
EAW-lex k=3	2.46	1.25	2.66	1.42	3.93	62%
EAW-pause k=2	3.16	1.51	3.05	1.03	4.28	65%

Policy	AL words	AL decision s	AL speech s	First out s	End offset s	Entities safe = simulated acc.
EAW-pause k=3	4.05	1.84	3.40	1.42	4.56	73%
sentence-by-sentence	6.92	2.77	5.80	4.51	7.94	97%



Findings. (i) Raising k trades latency for safety monotonically (wait-3 → wait-7: 54% → 76%). The k = 1–10 sweep's Pareto knee is wait-6 (alternatives wait-5, wait-7); the knee is weak (smaller than one entity), see the submission/ report. (ii) EAW helps exactly where its trigger is audible: self-corrections 0/4 under wait-3 vs 2/4 (EAW-lex) and 3/4 (EAW-pause). (iii) It does **not** help negation (5/11 vs 5/11): English commits 'not' before the Hindi clause-final nahin + verb is spoken, so no heard word can trigger the wait. This falsifies the purely source-triggered version of our idea; the wait must be triggered by the *target* (do not commit polarity or tense before the clause-final verb group). (iv) Speech-onset AL exceeds decision AL for every policy because English speech queues behind itself; extra waiting raises End Offset, which Sarvam scores. (v) Plain wait-k drifts when the English is shorter than the Hindi (clip01: 51 vs 68 words), so late English words run far ahead of their evidence (clip01 negation committed 16 Hindi words too early under wait-3). **Where to wait:** after a number until the scale word and 2 more words (catches 'nahin nahin, …'), and before English polarity until the Hindi verb group ends; elsewhere, speak early. **Cannot conclude:** anything about a real model's accuracy, statistical significance (6 clips, 1 speaker, 37 entities), or second-based numbers until real timings replace placeholders.

References

[1] Sarvam AI, Inter IIT 15.0 problem statement (local) and mini PS.
[2] Ma et al., STACL: Simultaneous Translation with Implicit Anticipation and Controllable Latency…, ACL 2019, 3025–3036.
[3] Ma et al., SimulEval: An Evaluation Toolkit for Simultaneous Translation, EMNLP 2020 Demos, 144–150; latency_scorer.py.
[4] Labiausse et al., High-Fidelity Simultaneous Speech-To-Speech Translation (Hibiki), ICML 2025, PMLR 267.
[5] Javed et al., IndicVoices, arXiv:2403.01926 (2024); HF card ai4bharat/IndicVoices; GitHub repo snapshot.
[6] AI4Bharat, IndicVoices-R HF card; Kathbath HF card.
[7] Papi et al., Over-Generation Cannot Be Rewarded: LAAL, AutoSimTrans 2022, 12–17.
[8] MahmoudAshraf/mms-300m-1130-forced-aligner (HF port of torchaudio MMS_FA); Pratap et al., MMS, 2023.
[9] WhisperX alignment.py; theainerd/Wav2Vec2-large-xlsr-hindi model card.
[10] torchaudio, Forced Alignment with Wav2Vec2 tutorial (local notebook). Full table: report/RESEARCH_PROPOSALS.md, SOURCE_NOTES.md.